



4주차 가이아 (1)

담당교수 임지연

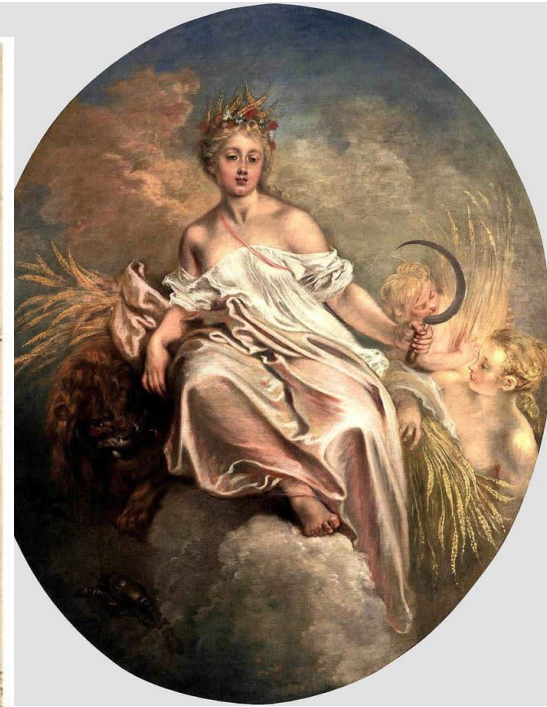
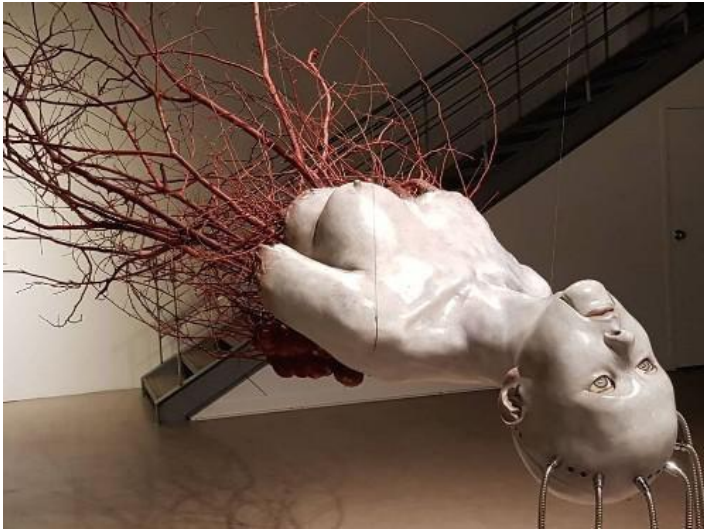
수업 목표

1. 제임스 러브록의 '가이아' 개념을 파악한다.
2. 러브록의 가이아 개념이 갖는 장점과 한계, 의미를 이해한다.

가이아 Gaia

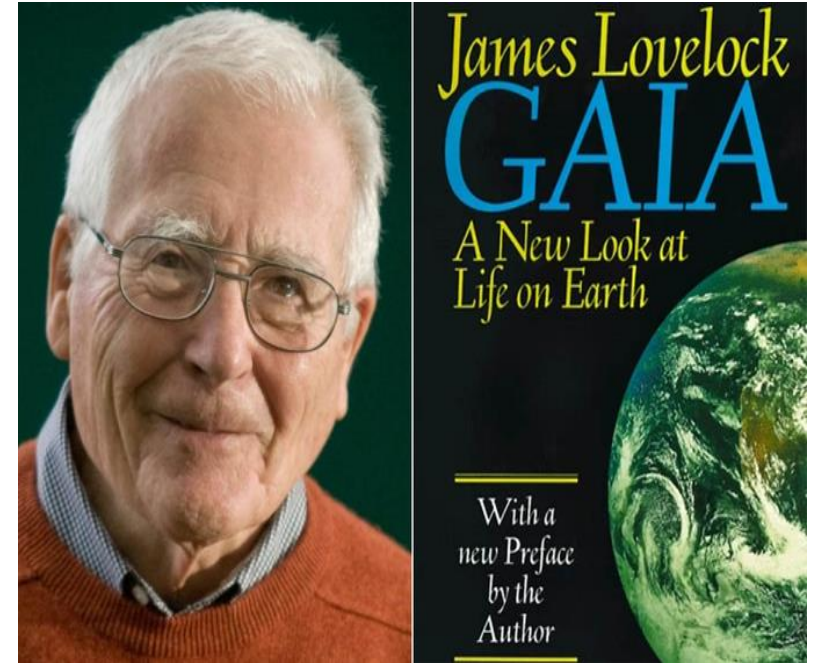


- 대지의 여신
- 생식과 창조, 생명을 관장하는 신
 - 자애롭고 너그러운 모성
 - 추상적이고 자연적인 힘
- 대지의 여신은 어느 문화권 신화에도 존재함



1. 가이아 개념 탄생 배경

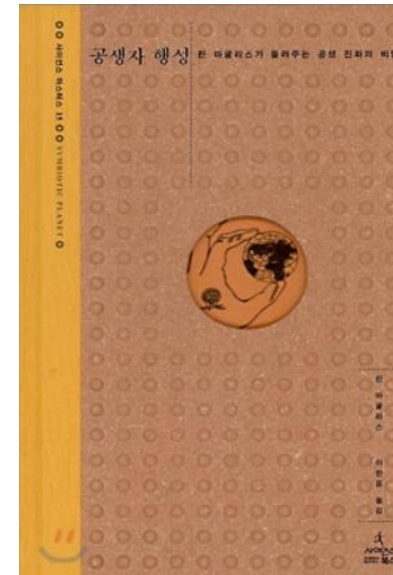
1) 제임스 러브록 James Ephraim Lovelock
(1919~2022. 영국 과학자, 의학 박사, 생물물리학 박사,
저술가, 발명가)



- 가이아 학설 창시
- 전자포획검출기(가스 성분 검출기), NASA 탐사 장비 발명
- 독립적 과학자

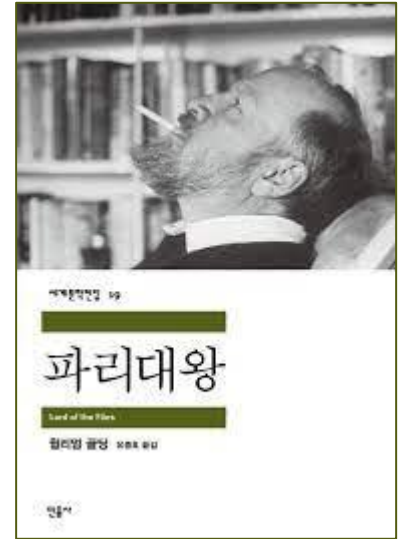
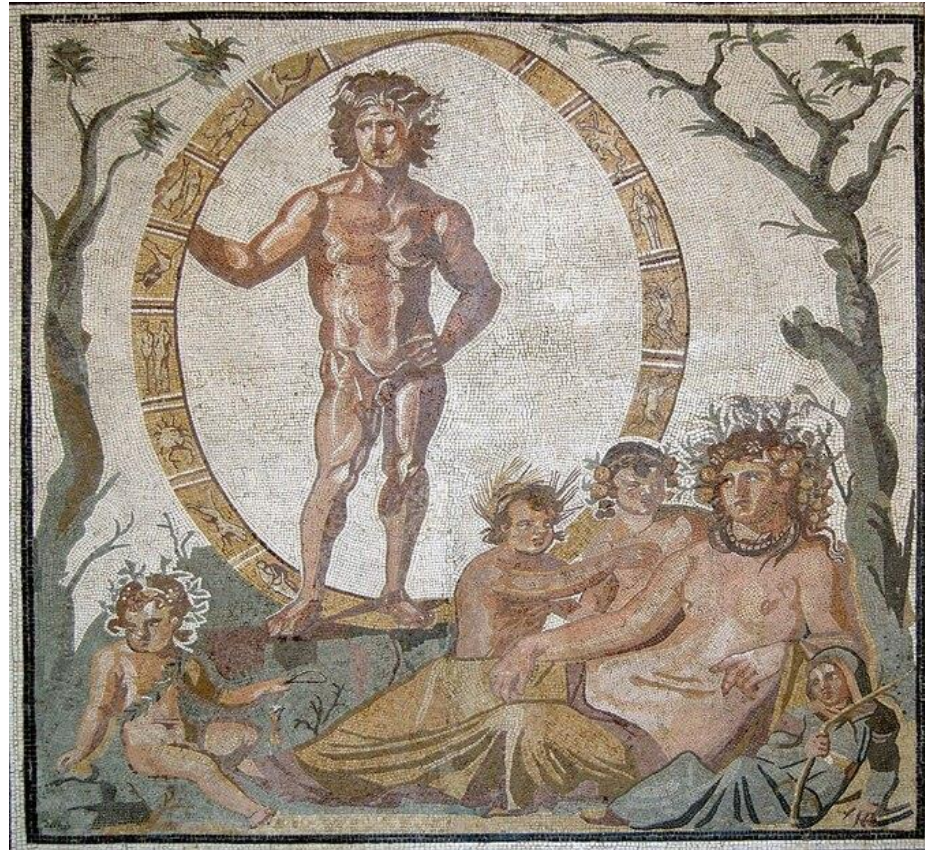
린 마굴리스 Lynn Margulis

- (1938~ 2011. 미국 생물학자. 세포생물학, 미생물 진화, 지구시스템 과학)
- NASA 지구생물학 연구 참여
- 세포내 공발생론(Endosymbiosis) 창설
- 러브록의 가이아 이론 지지
(칼 세이건과 결혼)



Gaia

윌리엄 골딩의 제안 고대 그리스 대지의 여신 이름



Atmospheric homeostasis by and for the biosphere: the gaia hypothesis

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(Manuscript received May 8; revised version August 20, 1973)

ABSTRACT

During the time, 3.2×10^9 years, that life has been present on Earth, the physical and chemical conditions of most of the planetary surface have never varied from those most favourable for life. The geological record reads that liquid water was always present and that the pH was never far from neutral. During this same period, however, the Earth's radiation environment underwent large changes. As the sun moved along the course set by the main sequence of stars its output will have increased at least 30 % and possibly 100 %. It may also have fluctuated in brightness over periods of a few million years. At the same time hydrogen was escaping to space from the Earth and so causing progressive changes in the chemical environment. This in turn through atmospheric compositional changes could have affected the Earth's radiation balance. It may have been that these physical and chemical changes always by blind chance followed the path whose bounds are the conditions favouring the continued existence of life. This paper offers an alternative explanation that, early after life began it acquired control of the planetary environment and that this homeostasis by and for the biosphere has persisted ever since. Historic and contemporary evidence and arguments for this hypothesis will be presented.

It is widely believed that the abundance of the principal gases N_2 and O_2 is determined by equilibrium chemistry. One of the larger problems in the atmospheric sciences is that of reconciling this belief with the uncomfortable fact that these same gases are cycled by the Biosphere with a geometric mean residence time measured in thousands of years. The more thoroughly the inventory of an individual gas is audited the more certain it seems that inorganic equilibrium or steady state processes determine its atmospheric concentration but the same audit frequently further reveals the extent of its biological involvement. A lucid account of contemporary information on the problem of the cycle of gases is in the paper of Junge (1972).

This paper presents a new view of the atmosphere, one in which it is seen as a component part of the biosphere rather than as a mere environment for life. In this new context the incompatibilities of biological cycles and in-

organic equilibria are seen as more apparent than real.

A starting point is a consideration of the profoundly anomalous composition of the Earth's atmosphere when it is compared with that of the expected atmosphere of a planet interpolated between Mars and Venus. Thus on Earth the simultaneous presence of O_2 and CH_4 at the present concentrations is a violation of the rules of equilibrium chemistry of no less than 30 orders of magnitude. Indeed so great is the disequilibrium among the gases of the Earth's atmosphere that it tends towards a combustible mixture, whereas the gases of Mars and Venus are close to chemical equilibrium and are more like combustion products.

The anomalous nature of the atmosphere has been known since Lewis & Randall (1923) first commented that at the pE and pH of the Earth the stable compound of nitrogen is the NO_3^- ion in the oceans; gaseous nitrogen should not be present. In spite of reminders by Hutchinson

Tellus XXVI (1974), 1-2

cludes the atmosphere and his boundary is therefore his skin. It might seem pointless therefore on Earth to seek the existence of a general living system, Gaia, in terms of entropy reductions within the atmosphere which clearly for some species is a sink for degraded products and energy but this neglects the fact that photosynthetic life uses premium radiation direct from the sun to sustain a high chemical potential gradient within the atmosphere on a planetary scale. For a tree, the boundary within which entropy is reduced is not its surface in contact with the atmosphere but rather the interface between the sun with the atmosphere as an extension of the tree. The tree produces not only food for consumers but also the equally important gas, oxygen which does not accumulate within the tree waiting to be eaten.

When the whole assembly of life is so seen it is clear that the true boundary is space. The outgoing entropy flux from the Earth indeed from Gaia 'if she exists', is long wavelength infra red radiation to space. This then, is the physical justification for delineating the boundary of life as the outer reaches of the atmosphere. There is also to a lesser extent an inner boundary represented by the interface with those inner parts of the Earth as yet unaffected by surface processes. We may now consider all that is encompassed by the bounds as putative life. Whether or not Gaia is real will depend upon the extent to which the entropy reduction within a compartment such as the atmosphere is recognisably different from the abiological steady state background.

On the matter of recognition a debt is owed to the fertile concept of information theory, Shannon & Weaver (1963). It has been demonstrated, for example by Evans (1969) that the classical properties, entropy and free energy, have exact information theoretic equivalents. Thus the information (I) of a system can be defined as

$$I = S_0 - S$$

where S_0 is the entropy of the components of the system at thermodynamic equilibrium and S the entropy of the system assembled. This relationship can be transferred directly from information theoretic to classical thermodynamic terms as follows:

$$I = (E + PV - TS - \sum N_i N_i)/T$$

Table 1. *The partial pressures in millibars of CO_2 , N_2 and O_2 on Mars, Earth and Venus and on two model abiological Earth's*

Model (D) after life had been deleted and Model (I) a geometric mean interpolation between Mars and Venus

Gas	Venus	Earth	Mars	Abiological earth models	
				D	I
CO_2	90 000	0.3	5	0.3-1 000	300
N_2	1 000	780	0.05	1	30
O_2	0	210	0.1	1	0.3

where the right hand side of the equation expresses information in terms of temperature (T) pressure (P) internal energy (E), volume (V), entropy (S), and the chemical potential (N) of the molecules present; it follows that information is a measure of disequilibrium in the classic sense and recognisability in the information theoretic sense.

By examining the extent to which the atmosphere is in chemical and physical disequilibrium both within itself and with the surface of the Earth we have a measure of the extent to which it is recognisable as a separate identity against a neutral background equilibrium state. Whether or not it is seen to be a component part of Gaia will depend upon the size of the disequilibrium revealed.

Thermodynamic evidence

We can determine the degree of departure from a conceivable abiological steady state atmosphere of the Earth by two means; firstly, by comparing the Earth's atmosphere with Mars and Venus which may be taken as representative of lifeless planets and secondly by comparing the present atmosphere with the atmosphere to be expected if life were deleted from Earth.

Table 1 shows the atmospheric composition of Mars and Venus compared with that of a hypothetical abiological Earth interpolated between them. Also shown is the atmospheric composition to be expected of the Earth if life were deleted. This was derived as follows: nitrogen reacts under the influence of solar

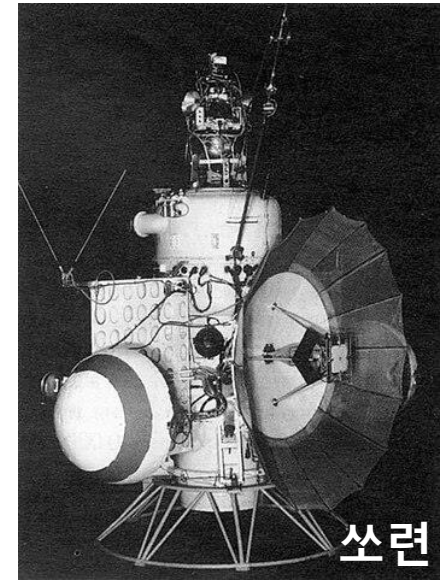
Tellus XXVI (1974), 1-2

• 러브록의 가이아 가설 목표

- 1) 지구 생물들에 감탄하는 사람들을 위한 가설
- 2) 자연을 정복 대상으로 보는 독선적 견해에 대한 대안
- 3) 행성 지구를 목적없이 태양계 지구를 도는 '애달픈 우주선' 으
로 표현하는 비관적 견해에 대한 대안

2) 제임스 러브록, 우주(외부)에서 지구를 내려다보다

내가 이 글을 쓰고 있는 지금 두 대의 바이킹 호 우주선이 지구로부터의 착륙 명령을 기다리면서 우리의 자매 행성 화성의 주위를 맴돌고 있다. 우주선들의 임무는 화성에 현존하거나 과거 한때 존재했던 생물들 또는 그 흔적으로 찾는 것이다.



소련 마스 1호
(1962년) 발사
성공

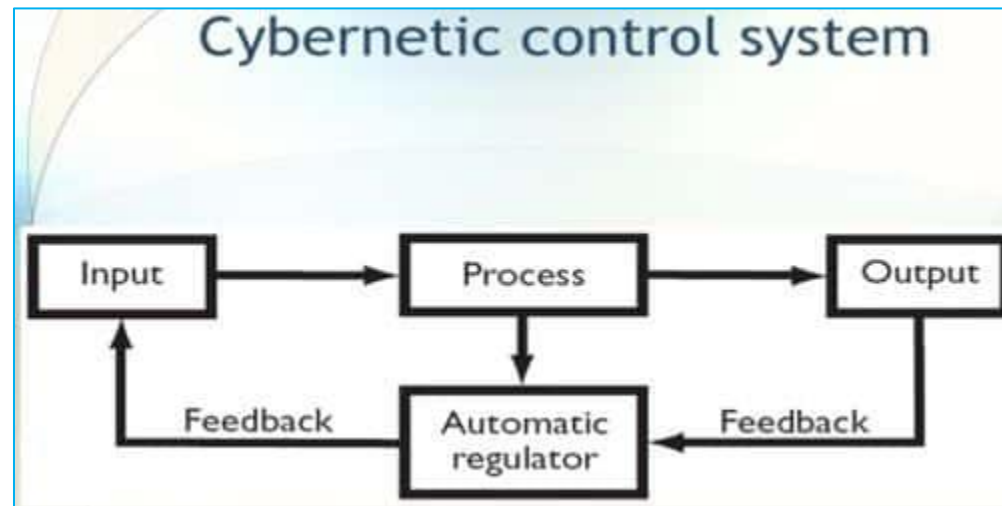


미국 매리너 4호, 화성
탐사 성공(1965)

2. 사이버네틱 시스템(Cybernetic system) 으로서의 가이아

- 활성화된 가이아
- 거대한 자기조절 시스템 huge self-regulation system
- 가이아는 살아있다

- 가이아는 신화적 비유보다 사이버네틱스 시스템
-자기조절 시스템
- 사이버네틱스 cybernetics:
'키잡이'를 뜻하는 그리스어 '쿠베르네테스'(kubernētēs, κυβερνήτης)에서 따옴.
'쿠베르네'는 '제어, 통제(govern)'를 의미



- 유기체와 기계를 외부 환경의 피드백을 통해 개체의 **항상성**을 유지하는 시스템
- 동물, 기계도 개체의 생존과 유지를 위해 외부 환경으로부터 정보를 받아들이고 이에 대한 피드백을 산출하는 일종의 정보처리 기계라는 점에서 인간과 다르지 않다는 통찰.

표. 생물이 없는 상상의 지구와 생물이 사는 현실 지구의 대기 성분 비교

구분	행성 이름			
	금성	생물이 없는 상상의 지구	화성	현실의 지구
이산화탄소	98%	98%	95%	0.03%
질소	1.9%	1.9%	2.7%	78%
산소	극미량	극미량	0.13%	21%
지표면 온도(°C)	477	290±50	-53	13
대기 압력(기압)	90	60	0.064	1.0

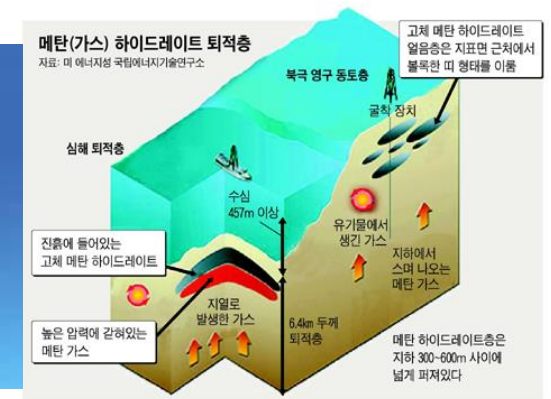
-제임스 러브록, 《가이아-살아있는 생명체로서의 지구》

• 기묘한 지구의 21% 산소 (O2)



메탄 가스 CH_4 가 산소 농도 규제

1) 성층권에서 2) 대기권에서



- 가이아의 정의: 생물군에 영향을 주고 영향받는 모든 것을 포함하는 행성적 생명 시스템. 모든 살아있는 유기체들과 항상성을 위한 능력을 공유하는 자기조절적 시스템 (self-regulating system)
- 가이아로서의 지구 또한 환경과 유기체들이 피드백을 주고받으며 상호작용하는 일종의 사이버네틱 시스템
- 가이아도 음과 양의 복잡한 피드백 작용을 통해 엔트로피를 감소시켜 항상성을 유지한다는 점에서 '살아있다'고 말할 수 있고, 이를 위해 정보를 받아들이고 처리한다는 점에서 제한적 수준이나마 '지능'을 갖고 있다.
- 생명이나 지능의 제한된 인간중심적 정의를 확장할 것을 제안
- 사이버네틱스 시스템의 비유- 생명, 지능을 사이버네틱스의 관점에서 설명함으로써 생명과 비생명, 인간과 기계 간의 단절보다 연속성 강조

가이아 이론에 대한 반응과 논쟁

- 1) 이론이 아니라 가설-신화적 이야기
- 2) 목적성에 대한 비판
 - 유사 과학(시적 과학)
- 3) 여신이자 어머니로 의인화-환경운동 기폭제
 - 페미니즘 비판